Introductory System

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Digital System: A digital system is a combination of devices designed to manipulate logical information or physical quantities that are represented in digital form.

Some popular digital system are:

* Digital computers
* Calculators
* Digital audio and video equipment
* Telephone system.

Analog System: Analog system contains devices that manipulate physical quantities that are represented in analog form.

Some common ~~ata~~ analog system are:

* Amplifiers
* Magnetic tape recording and Playbook
* equipment.
* Simple light dimmer switch

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2. Advantage of Digital Techniques over analog:

The advantages of digital techniques are gives below-

* Digital system are generally easier to design.
* Information storage is easy.
* Accuracy and precision are easier to maintain throughout the system.
* Operation can be programmed.
* Digital circuit are less affected by noice.
* More digital circuitry can be fabricated on IC chips.

To take advantages of digital tec-

~~Technique~~ Techniques: When dealing with analog inputs and outputs. Four steps must be followed –

* Convert the physical variable to an electrical signal (analog).
* Convert the electrical (analog) signal in to digital form.
* Process (operate on) the digital information.
* Convert the digital outputs back to real world analog form.

3. Convert binary to decimal $(1011.101)_2 = (?)_{10}$

Ans:

$$(1011.101)_2 = 1\times2^3 + 0\times2^2 + 1\times2^1 + 1\times2^0 + 1\times2^{-1} + 0\times2^{-2} + 1\times2^{-3}$$

$$= 8 + 0 + 2 + 1 + 0.5 + 0 + 0.125$$

$$= (11.625)_{10}$$

$(1011.101)_2 = (11.625)_{10}$ A.

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4. What is the largest numbers that can be represented using Eight bits.

Ans:

$$2^N - 1 = 2^8 - 1$$
$$= (255)_{10}$$
$$= (11111111)_2$$

5. What is the largest decimal value that can be represented using 12 bits.

Ans:

$$2^N - 1 = 2^{12} - 1$$
$$= (4095)_{10}$$

6. Describe the organization of a digital computer with figure.

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A computer is a System of hardware that performs arithmatic operation, manipulet data (usually in binary form) and makes decision thus computer is faster and more accurate.

Central Processing unit (cpu)

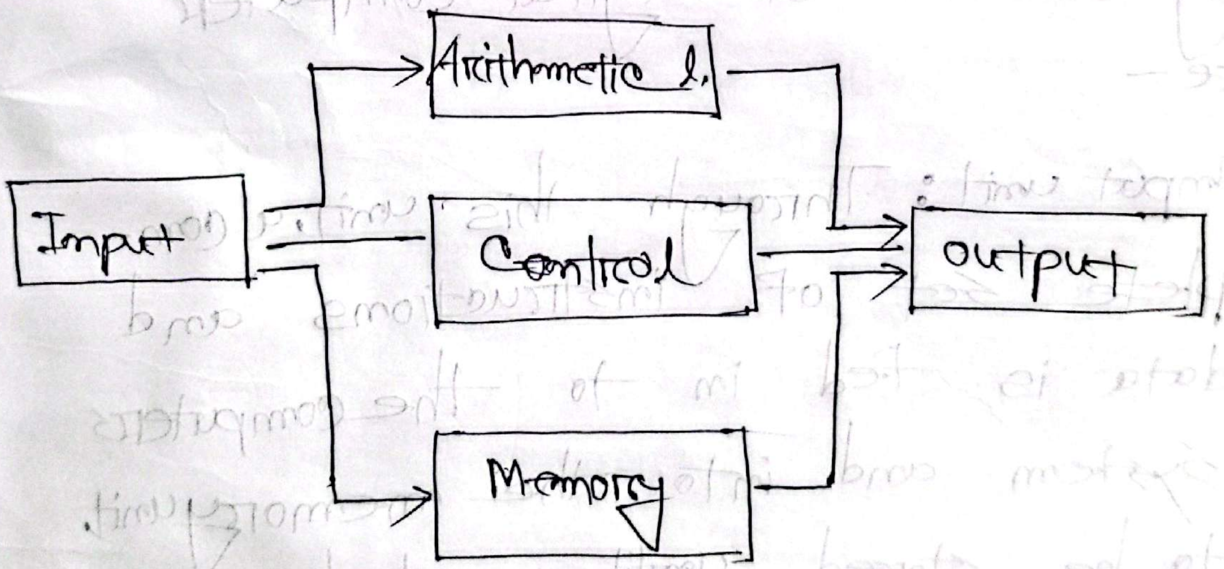

Fig: Diagram of digital computers.

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Ex: A human can take a list of 10 num and find the sum all in one operation by listing the number one over the others and adding team column by column. On the other hand it can add num only two at a time.

Organization of digital computer are -

Input unit: Through this unit, a complete set of instructions and data is fed in to the computers system and into the memory unit to be stored until needed. The information typically enters the input units. From a keyboard or a disk.

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Memory unit: The memory stores instr-uctions and Data unit from the input unit. It stores the isult of arithma-tic operations received from the rithmatic units. It also sypply the information the output units.

Control unit: This unit takes instructions frounet the memory unit once at a time. ~~and~~ inpea The core Processor that executes instructions. The control unit (CU) manages component operations. The ALU handles mathematical and logical calculations, and Registers Provide ultra-fast temporary storage for Processing data.

output unit: Converts Processed binary results from the computer into human

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readable formats for users.(e.g, monitor, Printer).

7. Convert the following operations:

$(526.56)_{10} = (?)_2 = (?)_8 = (?)_{16}$

Binary:

2	526	
2	263	0
2	131	1
2	65	1
2	32	1
2	16	0
2	8	0
2	4	0
2	2	0
2	1	0
	0	1

$.56 \times 2 = 1.12$

$.12 \times 2 = .24$

$.24 \times 2 = .48$

$.48 \times 2 = 0.96$

$.96 \times 2 = 1.92$

$.92 \times 2 = 1.84$

.56
×2
1.12
×2
0.24
×2
0.48
×2
0.96
×2
1.84
×2
1.68
×2
1.36
×2
0.72
×2
1.44
×2
0.88

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$(526.56)_{10} = (1000001110.100011...)$

Octal:

$$\begin{array}{r|l} 8 & 526 \\ 8 & 65 - 1 \\ 8 & 8 - 1 \\ 8 & 1 - 0 \\ & 0 - 1 \end{array}$$

$(526.56)_{10} =$
$(1011.4365560...)_8$

.56
×8
4.48
×8
3.84
×8
6.72
×8
5.76
×8
6.08
×8
0.64
×8
5.12
×8
0.96
×8
7.68
×8
5.44

Hexa:

$$\begin{array}{r|l} 16 & 526 \\ 16 & 32 - 1 \\ 16 & 2 - 0 \\ & 0 - 2 \end{array}$$

.56
×16
8.96
×16
15.36
×16
5.76
×16
12.16
×16
2.56
×16
8.96

$\therefore (526.56)_{10} = (001.8F5C28C...)_{16}$

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